

Analytical Merging Error for Gaussian Splats: An L^2 Framework for Optimal Approximation

Supplementary Material — Luxar Project

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Abstract

When simplifying a Gaussian-splat scene, a fundamental question arises: *how much information is lost when two splats are merged into one?* We derive a fully closed-form answer within the L^2 integrated squared error (ISE) framework. The key building block is the analytical inner product between any two Gaussians, which we combine with moment-matching to obtain a single-Gaussian approximation whose error is expressible via four distinct evaluations of one universal kernel (six terms total, with self-products as a special case). We show that the normalised merging error equals $\sin^2\alpha$, where α is the angle between the mixture and its best Gaussian approximation in L^2 . We provide geometric intuition, special-case simplifications, a cheaper ranking proxy based on the inter-splat cosine similarity, and pseudocode for practical implementation.

Contents

1	Introduction and Motivation	2
2	The Gaussian L^2 Inner Product	3
2.1	Setup and Notation	3
2.2	Derivation of the Kernel	3
3	The Moment-Matched Approximation	5
3.1	Why Moment-Matching?	5
3.2	Closed-Form Parameters	5
4	The Closed-Form Merging Error	6
4.1	Error Decomposition	6
4.2	Amplitude Optimisation	7
4.3	The Mergeability Score	7
4.4	Expanding the Score	8
5	Cheaper Proxy: The Inter-Splat Cosine	8

6	Special Cases	9
6.1	Equal Isotropic Covariances	9
6.2	Co-located Splats ($\boldsymbol{\mu}_1 = \boldsymbol{\mu}_2$)	10
6.3	Proportional Splats ($\phi_1 \propto \phi_2$)	10
7	Algorithm	10
8	Optimality Discussion	12
8.1	Is This the True Minimum L^2 Error?	12
8.2	When Does Moment-Matching Equal L^2 -Optimal?	13
8.3	Relationship to Existing Merging Criteria	13
8.4	Practical Considerations	13
9	Summary	13

1 Introduction and Motivation

Gaussian splatting represents a scene as a collection of N oriented Gaussians (“splats”), each parameterised by a centre $\boldsymbol{\mu}$, an amplitude a , and a covariance matrix $\boldsymbol{\Sigma}$ [4]. For large scientific volumes, N can reach millions, making simplification essential for interactive visualisation and efficient storage.

The most natural simplification primitive is *merging*: replacing two nearby or similar splats with a single splat that approximates their combined density as closely as possible (Fig. 1). To decide *which* pairs to merge and to quantify the resulting quality loss, we need a principled, closed-form measure of the **merging error**—the discrepancy between the two-splat mixture $\phi_1 + \phi_2$ and the best single-splat approximation g .

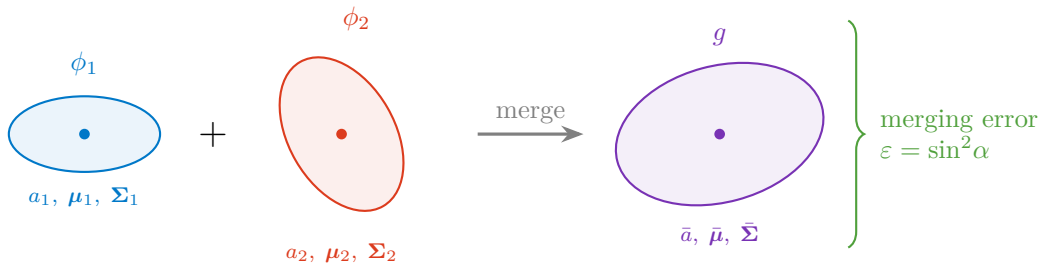


Figure 1: Schematic of the merging operation. Two oriented Gaussian splats ϕ_1 and ϕ_2 are replaced by a single splat g . The merging error ε quantifies the fraction of the mixture’s energy that cannot be captured by any single Gaussian.

Why L^2 ? Several divergences could measure approximation quality (Table 1). The L^2 (integrated squared error) framework is uniquely suited because *the integral of a product of two Gaussians has a closed-form expression*. Every term in the error expansion reduces to such a product, yielding a fully analytical result. By contrast, the Kullback–Leibler divergence for a Gaussian mixture

involves $\log(\phi_1 + \phi_2)$, which has no closed form [2]; the Hellinger distance requires $\sqrt{\phi_1 + \phi_2}$, equally intractable; and while the 2-Wasserstein distance between two individual Gaussians is elegant [3], it does not directly yield the error of approximating a *mixture*.

Table 1: Comparison of divergence measures for the merging problem. “Mixture error” refers to computing $d(\phi_1 + \phi_2, g)$ analytically.

Divergence	Between two Gaussians	Mixture error	Optimal g
L^2 (ISE)	Closed-form	Closed-form	Numerical [†]
KL	Closed-form	No closed-form	Moment-matched [‡]
Hellinger	Closed-form	No closed-form	No closed-form
W_2 (Wasserstein)	Closed-form	No closed-form	Bures barycenter

[†] Amplitude is closed-form; centre and covariance require a few Newton iterations.

[‡] Minimises $\text{KL}(f\|g)$ but the KL value itself has no closed form.

Outline. Section 2 derives the Gaussian L^2 inner product. Section 3 constructs the moment-matched single-Gaussian approximation. Section 4 states the closed-form error and the $\sin^2\alpha$ mergeability score. Section 5 introduces a cheaper ranking proxy. Section 6 treats important special cases. Section 7 provides pseudocode. Section 8 discusses optimality and practical considerations.

2 The Gaussian L^2 Inner Product

2.1 Setup and Notation

We work in $\mathbb{R}^D$. A **Gaussian splat** is a function

$$\phi(\mathbf{x}) = a \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})\right), \quad a > 0, \boldsymbol{\mu} \in \mathbb{R}^D, \boldsymbol{\Sigma} \in \mathbb{R}^{D \times D} \text{ s.p.d.} \quad (1)$$

The covariance $\boldsymbol{\Sigma}$ is stored via its lower-triangular Cholesky factor $\mathbf{L}$ satisfying $\boldsymbol{\Sigma} = \mathbf{L}\mathbf{L}^\top$. We denote the precision matrix $\mathbf{P} = \boldsymbol{\Sigma}^{-1}$. In the Luxar implementation, $\mathbf{L}$ is stored in packed row-major lower-triangular format with $D(D+1)/2$ elements.

The L^2 **inner product** and norm used throughout are

$$\langle f, g \rangle_{L^2} = \int_{\mathbb{R}^D} f(\mathbf{x}) g(\mathbf{x}) d\mathbf{x}, \quad \|f\|_{L^2}^2 = \langle f, f \rangle_{L^2}. \quad (2)$$

Since $a > 0$ and $\boldsymbol{\Sigma} \succ 0$, every splat has finite energy $\|\phi\|^2 < \infty$ and all inner products below are well-defined.

2.2 Derivation of the Kernel

Theorem 2.1 (Gaussian L^2 inner product). *For two splats ϕ_i, ϕ_j as in (1),*

$$\boxed{\langle \phi_i, \phi_j \rangle_{L^2} = a_i a_j (2\pi)^{D/2} \frac{\sqrt{|\boldsymbol{\Sigma}_i| |\boldsymbol{\Sigma}_j|}}{\sqrt{|\boldsymbol{\Sigma}_i + \boldsymbol{\Sigma}_j|}} \exp\left(-\frac{1}{2} \Delta \boldsymbol{\mu}_{ij}^\top (\boldsymbol{\Sigma}_i + \boldsymbol{\Sigma}_j)^{-1} \Delta \boldsymbol{\mu}_{ij}\right)} \quad (3)}$$

where $\Delta\boldsymbol{\mu}_{ij} = \boldsymbol{\mu}_i - \boldsymbol{\mu}_j$.

Proof. Write the product of the two Gaussian kernels:

$$\phi_i(\mathbf{x}) \phi_j(\mathbf{x}) = a_i a_j \exp(-\tfrac{1}{2}(Q_i + Q_j)), \quad (4)$$

where $Q_k = (\mathbf{x} - \boldsymbol{\mu}_k)^\top \boldsymbol{\Sigma}_k^{-1} (\mathbf{x} - \boldsymbol{\mu}_k)$. The sum of quadratic forms is itself quadratic in $\mathbf{x}$:

$$Q_i + Q_j = \underbrace{\mathbf{x}^\top (\boldsymbol{\Sigma}_i^{-1} + \boldsymbol{\Sigma}_j^{-1}) \mathbf{x}}_{\mathbf{P}_{ij}} - 2 \underbrace{\mathbf{x}^\top (\boldsymbol{\Sigma}_i^{-1} \boldsymbol{\mu}_i + \boldsymbol{\Sigma}_j^{-1} \boldsymbol{\mu}_j)}_{\boldsymbol{\eta}} + \underbrace{\boldsymbol{\mu}_i^\top \boldsymbol{\Sigma}_i^{-1} \boldsymbol{\mu}_i + \boldsymbol{\mu}_j^\top \boldsymbol{\Sigma}_j^{-1} \boldsymbol{\mu}_j}_c. \quad (5)$$

Completing the square:

$$Q_i + Q_j = (\mathbf{x} - \boldsymbol{\mu}_{ij})^\top \mathbf{P}_{ij} (\mathbf{x} - \boldsymbol{\mu}_{ij}) + (c - \boldsymbol{\eta}^\top \mathbf{P}_{ij}^{-1} \boldsymbol{\eta}), \quad (6)$$

where $\boldsymbol{\mu}_{ij} = \mathbf{P}_{ij}^{-1} \boldsymbol{\eta}$ is the “merged mean” in precision space. Integrating the Gaussian part over $\mathbb{R}^D$:

$$\int_{\mathbb{R}^D} \exp(-\tfrac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_{ij})^\top \mathbf{P}_{ij} (\mathbf{x} - \boldsymbol{\mu}_{ij})) d\mathbf{x} = (2\pi)^{D/2} |\mathbf{P}_{ij}^{-1}|^{1/2} = \frac{(2\pi)^{D/2}}{|\mathbf{P}_{ij}|^{1/2}}. \quad (7)$$

It remains to simplify the constant $c - \boldsymbol{\eta}^\top \mathbf{P}_{ij}^{-1} \boldsymbol{\eta}$. Using the identity (see [Lemma 2.2](#)):

$$c - \boldsymbol{\eta}^\top \mathbf{P}_{ij}^{-1} \boldsymbol{\eta} = \Delta\boldsymbol{\mu}_{ij}^\top \boldsymbol{\Sigma}_i^{-1} \mathbf{P}_{ij}^{-1} \boldsymbol{\Sigma}_j^{-1} \Delta\boldsymbol{\mu}_{ij} = \Delta\boldsymbol{\mu}_{ij}^\top (\boldsymbol{\Sigma}_i + \boldsymbol{\Sigma}_j)^{-1} \Delta\boldsymbol{\mu}_{ij}, \quad (8)$$

and the determinant (using the harmonic-mean identity $(\boldsymbol{\Sigma}_i^{-1} + \boldsymbol{\Sigma}_j^{-1})^{-1} = \boldsymbol{\Sigma}_i (\boldsymbol{\Sigma}_i + \boldsymbol{\Sigma}_j)^{-1} \boldsymbol{\Sigma}_j$, which follows from [Lemma 2.2](#) by taking the inverse of both sides of (10) after multiplying by $\boldsymbol{\Sigma}_i$ and $\boldsymbol{\Sigma}_j$):

$$|\mathbf{P}_{ij}^{-1}| = |(\boldsymbol{\Sigma}_i^{-1} + \boldsymbol{\Sigma}_j^{-1})^{-1}| = |\boldsymbol{\Sigma}_i (\boldsymbol{\Sigma}_i + \boldsymbol{\Sigma}_j)^{-1} \boldsymbol{\Sigma}_j| = \frac{|\boldsymbol{\Sigma}_i| |\boldsymbol{\Sigma}_j|}{|\boldsymbol{\Sigma}_i + \boldsymbol{\Sigma}_j|}. \quad (9)$$

Combining: the integral of (4) factors as $(2\pi)^{D/2} |\mathbf{P}_{ij}^{-1}|^{1/2}$ from (6) times $\exp(-\tfrac{1}{2} \cdot (8))$. Rewriting $|\mathbf{P}_{ij}^{-1}|$ via (9) yields (3). $\square$

Lemma 2.2. *For symmetric positive definite $\boldsymbol{\Sigma}_i, \boldsymbol{\Sigma}_j$:*

$$\boldsymbol{\Sigma}_i^{-1} (\boldsymbol{\Sigma}_i^{-1} + \boldsymbol{\Sigma}_j^{-1})^{-1} \boldsymbol{\Sigma}_j^{-1} = (\boldsymbol{\Sigma}_i + \boldsymbol{\Sigma}_j)^{-1}. \quad (10)$$

Proof. Multiply both sides on the left by $\boldsymbol{\Sigma}_i$ and on the right by $\boldsymbol{\Sigma}_j$. The left side becomes $(\boldsymbol{\Sigma}_i^{-1} + \boldsymbol{\Sigma}_j^{-1})^{-1}$ and the right side becomes $\boldsymbol{\Sigma}_i (\boldsymbol{\Sigma}_i + \boldsymbol{\Sigma}_j)^{-1} \boldsymbol{\Sigma}_j$. Taking the inverse of both sides: $\boldsymbol{\Sigma}_i^{-1} + \boldsymbol{\Sigma}_j^{-1}$ versus $\boldsymbol{\Sigma}_j^{-1} (\boldsymbol{\Sigma}_i + \boldsymbol{\Sigma}_j) \boldsymbol{\Sigma}_i^{-1} = \boldsymbol{\Sigma}_j^{-1} + \boldsymbol{\Sigma}_i^{-1}$. These are equal since matrix addition commutes. $\square$

Corollary 2.3 (Self inner product). *Setting $i = j$ and using $|2\boldsymbol{\Sigma}| = 2^D |\boldsymbol{\Sigma}|$:*

$$\|\phi_i\|_{L^2}^2 = a_i^2 \pi^{D/2} |\boldsymbol{\Sigma}_i|^{1/2}. \quad (11)$$

Remark 2.4 (In terms of Cholesky factors). Since $\boldsymbol{\Sigma}_i = \mathbf{L}_i \mathbf{L}_i^\top$, we have $|\boldsymbol{\Sigma}_i|^{1/2} = |\det \mathbf{L}_i| =$

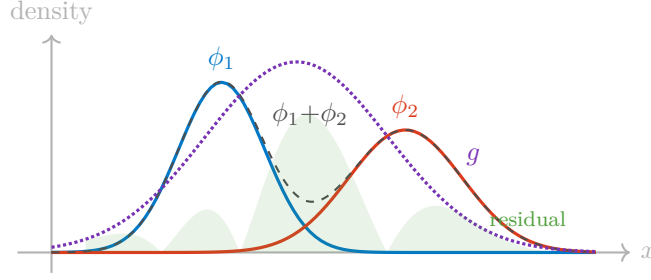


Figure 2: One-dimensional illustration. The mixture $\phi_1 + \phi_2$ (dashed) cannot be exactly represented by a single Gaussian g (dotted). The shaded area (schematic) represents the residual whose squared integral is the merging error E .

$\prod_{k=1}^D (\mathbf{L}_i)_{kk}$. To evaluate the kernel (3), form $\mathbf{S} = \mathbf{\Sigma}_1 + \mathbf{\Sigma}_2 = \mathbf{L}_1 \mathbf{L}_1^\top + \mathbf{L}_2 \mathbf{L}_2^\top$, compute $\mathbf{R} = \text{chol}(\mathbf{S})$, and then $|\mathbf{\Sigma}_1 + \mathbf{\Sigma}_2|^{1/2} = \prod_k \mathbf{R}_{kk}$. The Mahalanobis term $(\mathbf{\Sigma}_1 + \mathbf{\Sigma}_2)^{-1} \Delta \boldsymbol{\mu}$ is obtained by forward-backward substitution with $\mathbf{R}$, avoiding explicit matrix inversion.

3 The Moment-Matched Approximation

Given two splats ϕ_1, ϕ_2 , we seek a single splat $g(\mathbf{x}) = \bar{a} \bar{G}(\mathbf{x})$ that best approximates their sum $f = \phi_1 + \phi_2$. Throughout, we write $G_i(\mathbf{x}) = \exp(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_i)^\top \boldsymbol{\Sigma}_i^{-1}(\mathbf{x} - \boldsymbol{\mu}_i))$ for the **unit-amplitude kernel** of splat i , so that $\phi_i = a_i G_i$. Similarly, $\bar{G}(\mathbf{x}) = \exp(-\frac{1}{2}(\mathbf{x} - \bar{\boldsymbol{\mu}})^\top \bar{\boldsymbol{\Sigma}}^{-1}(\mathbf{x} - \bar{\boldsymbol{\mu}}))$ denotes the unit-amplitude template for the merged splat.

3.1 Why Moment-Matching?

The Gaussian family forms an exponential family with sufficient statistics $\mathbf{x}$ and $\mathbf{x}\mathbf{x}^\top$ [1]. A classical result from information geometry states that the minimiser of the forward KL divergence $\text{KL}\left(\frac{f}{\int f} \parallel \frac{g}{\int g}\right)$ over any member of an exponential family matches the sufficient statistics of the target—i.e., it matches the mean and covariance of the *normalised* mixture [1, 8].

Applied to our problem: we first normalise $f = \phi_1 + \phi_2$ to a probability distribution, find the Gaussian whose mean and covariance match (determining $\bar{\boldsymbol{\mu}}$ and $\bar{\boldsymbol{\Sigma}}$), and then set the amplitude $\bar{a}$ separately—either by mass conservation (Section 3.2) or by L^2 optimisation (Section 4.2). The moment-matched Gaussian is therefore the canonical choice for single-component approximation. Its shape parameters have fully closed-form expressions.

3.2 Closed-Form Parameters

Define the **mass** of each splat and the mass-based weights:

$$m_i = a_i (2\pi)^{D/2} |\boldsymbol{\Sigma}_i|^{1/2} = \int_{\mathbb{R}^D} \phi_i(\mathbf{x}) d\mathbf{x}, \quad w_i = \frac{m_i}{m_1 + m_2}. \quad (12)$$

Proposition 3.1 (Moment-matched parameters). *The moment-matched single Gaussian g has*

parameters:

$$\bar{\boldsymbol{\mu}} = w_1 \boldsymbol{\mu}_1 + w_2 \boldsymbol{\mu}_2, \quad (13)$$

$$\bar{\boldsymbol{\Sigma}} = w_1 \boldsymbol{\Sigma}_1 + w_2 \boldsymbol{\Sigma}_2 + w_1 w_2 \Delta \boldsymbol{\mu} \Delta \boldsymbol{\mu}^\top, \quad (14)$$

$$\bar{a} = \frac{m_1 + m_2}{(2\pi)^{D/2} |\bar{\boldsymbol{\Sigma}}|^{1/2}} = \frac{a_1 |\boldsymbol{\Sigma}_1|^{1/2} + a_2 |\boldsymbol{\Sigma}_2|^{1/2}}{|\bar{\boldsymbol{\Sigma}}|^{1/2}}, \quad (15)$$

where $\Delta \boldsymbol{\mu} = \boldsymbol{\mu}_1 - \boldsymbol{\mu}_2$.

Proof. Treat the mixture as an unnormalised density with total mass $M = m_1 + m_2$. Normalising: $p(\mathbf{x}) = w_1 \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}_1, \boldsymbol{\Sigma}_1) + w_2 \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}_2, \boldsymbol{\Sigma}_2)$. The mean is $\bar{\boldsymbol{\mu}} = w_1 \boldsymbol{\mu}_1 + w_2 \boldsymbol{\mu}_2$ by linearity. The covariance uses the law of total variance:

$$\bar{\boldsymbol{\Sigma}} = \sum_i w_i \boldsymbol{\Sigma}_i + \sum_i w_i (\boldsymbol{\mu}_i - \bar{\boldsymbol{\mu}})(\boldsymbol{\mu}_i - \bar{\boldsymbol{\mu}})^\top. \quad (16)$$

Substituting $\boldsymbol{\mu}_i - \bar{\boldsymbol{\mu}} = w_j (\boldsymbol{\mu}_i - \boldsymbol{\mu}_j)$ (where $j \neq i$) and $w_1 w_2^2 + w_2 w_1^2 = w_1 w_2$ yields (14). The amplitude (15) enforces mass conservation $\int g = M$. $\square$

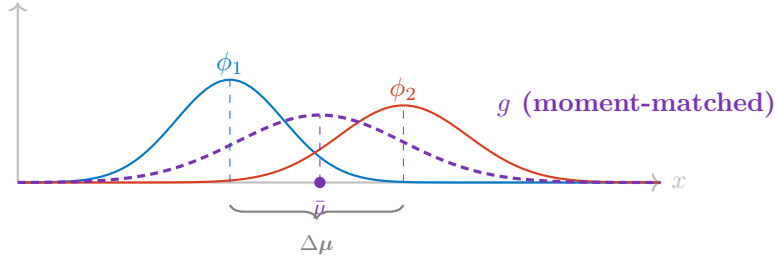


Figure 3: The moment-matched Gaussian g (purple dashed) has its mean $\bar{\boldsymbol{\mu}}$ at the mass-weighted average of the two centres, and its covariance includes a rank-1 “spreading” term $w_1 w_2 \Delta \boldsymbol{\mu} \Delta \boldsymbol{\mu}^\top$ due to the separation.

Remark 3.2 (Geometric interpretation of $\bar{\boldsymbol{\Sigma}}$). The moment-matched covariance decomposes as $\bar{\boldsymbol{\Sigma}} = \bar{\boldsymbol{\Sigma}}_{\text{intra}} + \bar{\boldsymbol{\Sigma}}_{\text{inter}}$, where $\bar{\boldsymbol{\Sigma}}_{\text{intra}} = w_1 \boldsymbol{\Sigma}_1 + w_2 \boldsymbol{\Sigma}_2$ is the weighted average of the individual spreads (the “intrinsic” width), and $\bar{\boldsymbol{\Sigma}}_{\text{inter}} = w_1 w_2 \Delta \boldsymbol{\mu} \Delta \boldsymbol{\mu}^\top$ is a rank-1 term encoding the “extrinsic” spread from the separation of the two centres. When $\boldsymbol{\mu}_1 = \boldsymbol{\mu}_2$, the extrinsic term vanishes.

4 The Closed-Form Merging Error

4.1 Error Decomposition

For *any* single-Gaussian approximation g , the L^2 error is:

$$E = \|\phi_1 + \phi_2 - g\|_{L^2}^2 = \underbrace{K_{11} + 2K_{12} + K_{22}}_{\|f\|^2} - 2K_{1g} - 2K_{2g} + K_{gg}, \quad (17)$$

where $K_{ij} = \langle \phi_i, \phi_j \rangle_{L^2}$ and all six terms use the kernel from [Theorem 2.1](#).

4.2 Amplitude Optimisation

Rather than using the mass-preserving amplitude $\bar{a}$ from (15), we can fix the moment-matched shape parameters $(\bar{\boldsymbol{\mu}}, \bar{\boldsymbol{\Sigma}})$ and choose the amplitude that minimises the L^2 error. Write $\bar{G}(\mathbf{x}) = \exp(-\frac{1}{2}(\mathbf{x} - \bar{\boldsymbol{\mu}})^\top \bar{\boldsymbol{\Sigma}}^{-1}(\mathbf{x} - \bar{\boldsymbol{\mu}}))$ for the unit-amplitude template. Since $E = \|f\|^2 - 2\bar{a} \langle f, \bar{G} \rangle + \bar{a}^2 \|\bar{G}\|^2$ is quadratic in $\bar{a}$, the minimiser is

$$a^* = \frac{\langle f, \bar{G} \rangle}{\|\bar{G}\|^2} = \frac{a_1 \langle G_1, \bar{G} \rangle + a_2 \langle G_2, \bar{G} \rangle}{\pi^{D/2} |\bar{\boldsymbol{\Sigma}}|^{1/2}}, \quad (18)$$

and the corresponding minimum error is the classical **projection residual**:

$$E^* = \|f\|^2 - \frac{|\langle f, \bar{G} \rangle|^2}{\|\bar{G}\|^2} \quad (19)$$

This is strictly better than (or equal to) the mass-preserving error, and still fully closed-form.

4.3 The Mergeability Score

Normalising (19) by the mixture's total energy gives a dimensionless score (Fig. 4):

$$\varepsilon = \frac{E^*}{\|f\|^2} = 1 - \frac{|\langle f, \bar{G} \rangle|^2}{\|f\|^2 \|\bar{G}\|^2} = 1 - \cos^2 \alpha = \sin^2 \alpha \quad (20)$$

where α is the **angle between f and $\bar{G}$ in L^2** (the Hilbert space of square-integrable functions).

- $\varepsilon = 0$ ($\alpha = 0$): the two splats are proportional ($\phi_1 \propto \phi_2$), and the merge is perfect.
- $\varepsilon \rightarrow 1$ ($\alpha \rightarrow \pi/2$): the mixture is nearly orthogonal to every single Gaussian with these shape parameters, and the merge is catastrophically lossy.

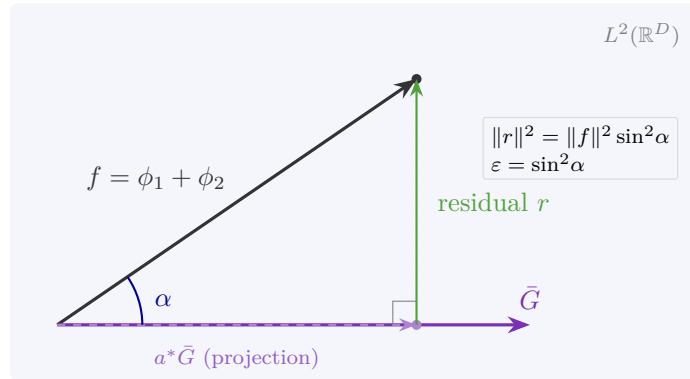


Figure 4: Geometric interpretation in L^2 . The mixture $f = \phi_1 + \phi_2$ is a vector in the Hilbert space. Its projection onto the ray spanned by $\bar{G}$ (the unit-amplitude template) gives the optimal amplitude $a^* \bar{G}$. The residual $r = f - a^* \bar{G}$ is orthogonal to $\bar{G}$, and the normalised squared residual is $\varepsilon = \sin^2 \alpha$.

4.4 Expanding the Score

Writing out all terms in (20) using the kernel (3):

$$\varepsilon = 1 - \frac{(a_1 I_1 + a_2 I_2)^2}{(a_1^2 J_1 + 2 a_1 a_2 I_{12} + a_2^2 J_2) \cdot J_g} \quad (21)$$

with the four kernel evaluations listed in Table 2.

Table 2: The four distinct kernel evaluations needed for the mergeability score. All use the master formula (3).

Symbol	Role	Expression
$J_i = \frac{\ \phi_i\ ^2}{a_i^2}$	Self-energy (per unit a_i^2)	$\pi^{D/2} \Sigma_i ^{1/2}$
$J_g = \ \tilde{G}\ ^2$	Template self-energy	$\pi^{D/2} \bar{\Sigma} ^{1/2}$
$I_{12} = \langle G_1, G_2 \rangle$	Cross-kernel (input splats)	$(2\pi)^{D/2} \frac{\sqrt{ \Sigma_1 \Sigma_2 }}{\sqrt{ \Sigma_1 + \Sigma_2 }} e^{-Q_{12}/2}$
$I_i = \langle G_i, \tilde{G} \rangle$	Cross-kernel (splat vs. template)	$(2\pi)^{D/2} \frac{\sqrt{ \Sigma_i \bar{\Sigma} }}{\sqrt{ \Sigma_i + \bar{\Sigma} }} e^{-\bar{Q}_i/2}$

$$Q_{12} = \Delta \mu^\top (\Sigma_1 + \Sigma_2)^{-1} \Delta \mu, \quad \bar{Q}_i = (\bar{\mu} - \mu_i)^\top (\Sigma_i + \bar{\Sigma})^{-1} (\bar{\mu} - \mu_i).$$

5 Cheaper Proxy: The Inter-Splat Cosine

Computing the full mergeability score requires four kernel evaluations and forming the moment-matched covariance. When the goal is merely to *rank* splat pairs by mergeability (e.g., to select the best pair to merge next), a simpler proxy suffices:

Proposition 5.1 (Inter-splat L^2 cosine). *The cosine similarity between two splats in L^2 is amplitude-independent:*

$$\cos \theta = \frac{\langle \phi_1, \phi_2 \rangle}{\|\phi_1\| \|\phi_2\|} = 2^{D/2} \frac{(|\Sigma_1| |\Sigma_2|)^{1/4}}{|\Sigma_1 + \Sigma_2|^{1/2}} \exp(-\tfrac{1}{2} Q_{12}) \quad (22)$$

where $Q_{12} = \Delta \mu^\top (\Sigma_1 + \Sigma_2)^{-1} \Delta \mu$.

Proof. Direct computation from (3) and (11): the amplitudes a_1, a_2 cancel in the ratio, and $(2\pi)^{D/2} / (\pi^{D/2})^{1/2+1/2} = 2^{D/2}$. $\square$

Properties.

- $\cos \theta = 1$ iff $\Sigma_1 = \Sigma_2$ and $\mu_1 = \mu_2$ (identical shape and location).
- $\cos \theta \rightarrow 0$ as $Q_{12} \rightarrow \infty$ (well-separated) or as the covariance shapes diverge.
- Requires only **one** kernel evaluation (plus trivial self-products).

- Strongly correlated with ε in practice, but does not account for amplitude ratios. (By contrast, the full mergeability score ε depends on the ratio a_1/a_2 through $\|f\|^2$.)

Remark 5.2 (Relationship between θ and α). The inter-splat angle θ (between ϕ_1 and ϕ_2) and the mergeability angle α (between the mixture f and $\bar{G}$) are distinct quantities. They agree in the extreme cases: $\cos \theta = 1 \Leftrightarrow \varepsilon = 0$ (proportional splats) and $\cos \theta = 0 \Rightarrow \varepsilon = 1$ (orthogonal splats). In between, the relationship is non-trivial because α depends on the moment-matched template $\bar{G}$, while θ is a direct pairwise measure. Nevertheless, $\cos \theta$ is an effective proxy for ranking pairs by mergeability.

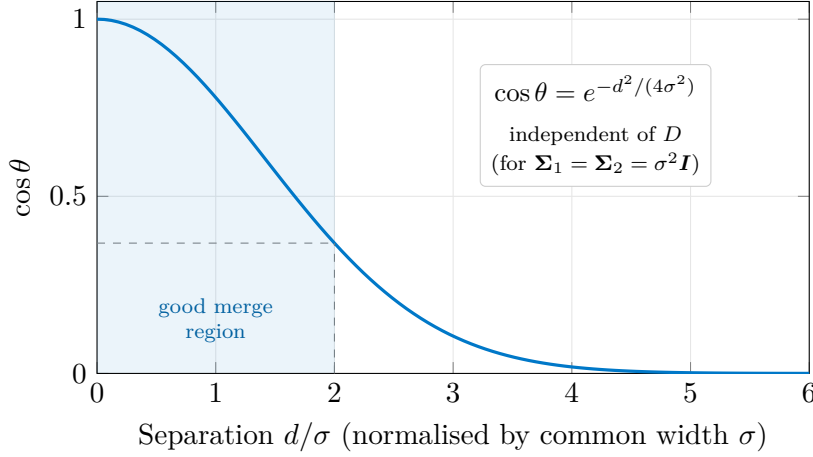


Figure 5: Inter-splat cosine similarity vs. normalised separation for two splats with equal isotropic covariances $\Sigma_1 = \Sigma_2 = \sigma^2 \mathbf{I}$. The result $\cos \theta = \exp(-d^2/4\sigma^2)$ is independent of the dimension D : the $2^{D/2}$ prefactor in (22) exactly cancels the determinant ratio for equal covariances. The shaded region ($d \lesssim 2\sigma$, $\cos \theta \gtrsim e^{-1}$) indicates practically mergeable pairs.

6 Special Cases

6.1 Equal Isotropic Covariances

When $\Sigma_1 = \Sigma_2 = \sigma^2 \mathbf{I}_D$, the kernel simplifies dramatically. Let $d = \|\Delta \mu\|$:

$$K_{12} = a_1 a_2 \pi^{D/2} \sigma^D e^{-d^2/(4\sigma^2)}, \quad (23)$$

$$\|f\|^2 = (a_1^2 + a_2^2) \pi^{D/2} \sigma^D + 2 a_1 a_2 \pi^{D/2} \sigma^D e^{-d^2/(4\sigma^2)}, \quad (24)$$

$$\cos \theta = e^{-d^2/(4\sigma^2)}. \quad (25)$$

The dimension-dependence vanishes because the $2^{D/2}$ prefactor in (22) exactly cancels the determinant ratio $|\sigma^2 \mathbf{I}|^{1/2}/|2\sigma^2 \mathbf{I}|^{1/2} = 2^{-D/2}$ for equal covariances. The inter-splat cosine thus depends only on the ratio d/σ , offering a simple and intuitive merging criterion: merge when $d \lesssim 2\sigma$ (where $\cos \theta \gtrsim e^{-1} \approx 0.37$).

6.2 Co-located Splats ($\mu_1 = \mu_2$)

When the centres coincide, $Q_{12} = 0$ and $\bar{\Sigma}_{\text{inter}} = 0$, so:

$$\cos \theta = 2^{D/2} \frac{(|\Sigma_1| |\Sigma_2|)^{1/4}}{|\Sigma_1 + \Sigma_2|^{1/2}}. \quad (26)$$

This depends purely on the *shape* discrepancy. For isotropic splats with $\Sigma_i = \sigma_i^2 \mathbf{I}$:

$$\cos \theta = \frac{2^{D/2} (\sigma_1 \sigma_2)^{D/2}}{(\sigma_1^2 + \sigma_2^2)^{D/2}} = \left(\frac{2\sigma_1 \sigma_2}{\sigma_1^2 + \sigma_2^2} \right)^{D/2} = \left(\frac{2}{r + r^{-1}} \right)^{D/2}, \quad (27)$$

where $r = \sigma_1/\sigma_2$ is the width ratio. This is the geometric-harmonic mean ratio raised to the $D/2$ power. It equals 1 when $r = 1$ and decays rapidly as r departs from unity, especially in high dimensions (Fig. 6).

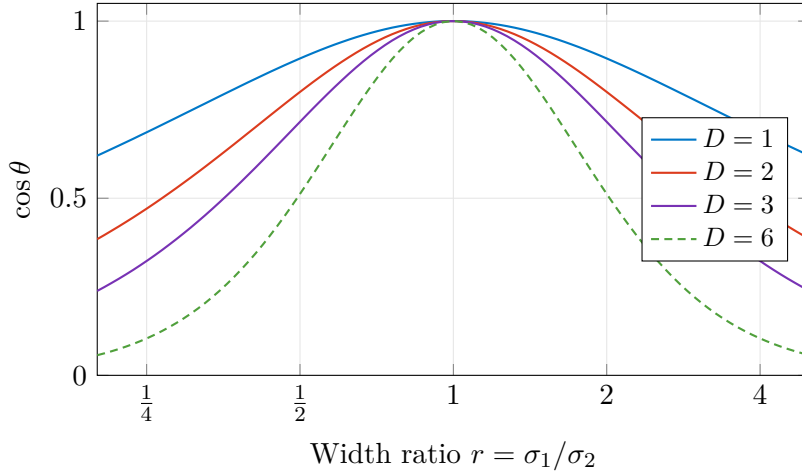


Figure 6: Inter-splat cosine for co-located isotropic splats as a function of the width ratio. Higher dimensions penalise shape differences more severely: in $D = 3$, a $2\times$ width ratio already drops the cosine to ≈ 0.60 .

6.3 Proportional Splats ($\phi_1 \propto \phi_2$)

When $\mu_1 = \mu_2$ and $\Sigma_1 = \Sigma_2$, the two splats are proportional: $\phi_1 + \phi_2 = (a_1 + a_2) G(\mathbf{x}; \mu, \Sigma)$, which is already a single Gaussian. In this case $\cos \theta = 1$ and $\varepsilon = 0$: the merge is exact.

7 Algorithm

[Algorithm 1](#) computes the full mergeability score for a pair of splats. [Algorithm 2](#) provides the cheaper cosine proxy for ranking. [Algorithm 3](#) constructs the merged splat using either mass-preserving or L^2 -optimal amplitude.

Algorithm 1: Mergeability score for a pair of Gaussian splats

Input: Splats $(a_1, \boldsymbol{\mu}_1, \mathbf{L}_1)$ and $(a_2, \boldsymbol{\mu}_2, \mathbf{L}_2)$ in $\mathbb{R}^D$

Output: Mergeability score $\varepsilon \in [0, 1]$ and optimal merged parameters $(\bar{a}, \bar{\boldsymbol{\mu}}, \bar{\boldsymbol{\Sigma}})$

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/* Step 1: Form covariances */
1  $\boldsymbol{\Sigma}_i \leftarrow \mathbf{L}_i \mathbf{L}_i^\top$  for  $i = 1, 2$ ;

/* Step 2: Moment-matched parameters */
2  $d_i \leftarrow \prod_k (\mathbf{L}_i)_{kk}$ ; //  $|\boldsymbol{\Sigma}_i|^{1/2} = |\det \mathbf{L}_i|$ 
3  $m_i \leftarrow a_i (2\pi)^{D/2} d_i$ ; // mass
4  $w_i \leftarrow m_i / (m_1 + m_2)$ ;
5  $\bar{\boldsymbol{\mu}} \leftarrow w_1 \boldsymbol{\mu}_1 + w_2 \boldsymbol{\mu}_2$ ;
6  $\Delta \boldsymbol{\mu} \leftarrow \boldsymbol{\mu}_1 - \boldsymbol{\mu}_2$ ;
7  $\bar{\boldsymbol{\Sigma}} \leftarrow w_1 \boldsymbol{\Sigma}_1 + w_2 \boldsymbol{\Sigma}_2 + w_1 w_2 \Delta \boldsymbol{\mu} \Delta \boldsymbol{\mu}^\top$ ;
8  $\bar{\mathbf{L}} \leftarrow \text{chol}(\bar{\boldsymbol{\Sigma}})$ ;
9  $d_g \leftarrow \prod_k \bar{\mathbf{L}}_{kk}$ ; //  $|\bar{\boldsymbol{\Sigma}}|^{1/2}$ 

/* Step 3: Self-energies */
10  $J_i \leftarrow \pi^{D/2} d_i$  for  $i = 1, 2$ ;
11  $J_g \leftarrow \pi^{D/2} d_g$ ;

/* Step 4: Cross-kernels (input pair) */
12  $\mathbf{S}_{12} \leftarrow \boldsymbol{\Sigma}_1 + \boldsymbol{\Sigma}_2$ ;
13  $\mathbf{R}_{12} \leftarrow \text{chol}(\mathbf{S}_{12})$ ;
14  $\mathbf{v} \leftarrow \text{solve } \mathbf{S}_{12} \mathbf{v} = \Delta \boldsymbol{\mu}$ ; // via  $\mathbf{R}_{12}$ 
15  $\mathbf{Q}_{12} \leftarrow \Delta \boldsymbol{\mu}^\top \mathbf{v}$ ;
16  $d_{12} \leftarrow \prod_k (\mathbf{R}_{12})_{kk}$ ;
17  $I_{12} \leftarrow (2\pi)^{D/2} (d_1 d_2) / d_{12} \cdot \exp(-\mathbf{Q}_{12}/2)$ ;

/* Step 5: Cross-kernels (splat vs. template) */
18 for  $i \in \{1, 2\}$  do
19    $\mathbf{S}_i \leftarrow \boldsymbol{\Sigma}_i + \bar{\boldsymbol{\Sigma}}$ ;
20    $\mathbf{R}_i \leftarrow \text{chol}(\mathbf{S}_i)$ ;
21    $\mathbf{z} \leftarrow \text{solve } \mathbf{S}_i \mathbf{z} = \bar{\boldsymbol{\mu}} - \boldsymbol{\mu}_i$ ;
22    $\bar{\mathbf{Q}}_i \leftarrow (\bar{\boldsymbol{\mu}} - \boldsymbol{\mu}_i)^\top \mathbf{z}$ ;
23    $I_i \leftarrow (2\pi)^{D/2} (d_i d_g) / \prod_k (\mathbf{R}_i)_{kk} \cdot \exp(-\bar{\mathbf{Q}}_i/2)$ ;

/* Step 6: Mergeability score */
24  $\|f\|^2 \leftarrow a_1^2 J_1 + 2 a_1 a_2 I_{12} + a_2^2 J_2$ ;
25  $\langle f, \bar{G} \rangle \leftarrow a_1 I_1 + a_2 I_2$ ;
26  $\varepsilon \leftarrow 1 - \langle f, \bar{G} \rangle^2 / (\|f\|^2 \cdot J_g)$ ;

/* Step 7:  $L^2$ -optimal amplitude */
27  $\bar{a} \leftarrow \langle f, \bar{G} \rangle / J_g$ ;
28 return  $\varepsilon, (\bar{a}, \bar{\boldsymbol{\mu}}, \bar{\boldsymbol{\Sigma}})$ ;

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Algorithm 2: Quick mergeability ranking via inter-splat cosine

Input: Splats $(a_1, \boldsymbol{\mu}_1, \mathbf{L}_1)$ and $(a_2, \boldsymbol{\mu}_2, \mathbf{L}_2)$

Output: Cosine similarity $\cos \theta \in [0, 1]$

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1  $\boldsymbol{\Sigma}_i \leftarrow \mathbf{L}_i \mathbf{L}_i^\top$  for  $i = 1, 2$ ;  
2  $\mathbf{S} \leftarrow \boldsymbol{\Sigma}_1 + \boldsymbol{\Sigma}_2$ ;  
3  $\mathbf{R} \leftarrow \text{chol}(\mathbf{S})$ ;  
4  $\Delta \boldsymbol{\mu} \leftarrow \boldsymbol{\mu}_1 - \boldsymbol{\mu}_2$ ;  
5  $\mathbf{w} \leftarrow \text{solve } \mathbf{S} \mathbf{w} = \Delta \boldsymbol{\mu}$ ;  
6  $Q \leftarrow \Delta \boldsymbol{\mu}^\top \mathbf{w}$ ;  
7  $\cos \theta \leftarrow 2^{D/2} \cdot \frac{(\prod_k (\mathbf{L}_1)_{kk})^{1/2} (\prod_k (\mathbf{L}_2)_{kk})^{1/2}}{\prod_k (\mathbf{R})_{kk}} \cdot e^{-Q/2}$ ;  
8 return  $\cos \theta$ ;
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Algorithm 3: Construct the merged splat

Input: Splats $(a_i, \boldsymbol{\mu}_i, \mathbf{L}_i)$ for $i = 1, 2$; desired mode (**mass** or **L2**)

Output: Merged splat $(\bar{a}, \bar{\boldsymbol{\mu}}, \bar{\mathbf{L}})$

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1 Compute  $\bar{\boldsymbol{\mu}}, \bar{\boldsymbol{\Sigma}}, \bar{\mathbf{L}} = \text{chol}(\bar{\boldsymbol{\Sigma}})$  as in Algorithm 1 Steps 1–2;  
2 if mode = mass then  
3    $m_i \leftarrow a_i (2\pi)^{D/2} \prod_k (\mathbf{L}_i)_{kk}$  for  $i = 1, 2$ ;  
4    $\bar{a} \leftarrow (m_1 + m_2) / ((2\pi)^{D/2} \prod_k \bar{\mathbf{L}}_{kk})$  ; // mass-preserving  
5 else  
6   Compute  $I_1, I_2$  and  $J_g$  as in Algorithm 1 Steps 3–5;  
7    $\bar{a} \leftarrow (a_1 I_1 + a_2 I_2) / J_g$  ; //  $L^2$ -optimal  
8 return  $(\bar{a}, \bar{\boldsymbol{\mu}}, \bar{\mathbf{L}})$ ;
```

8 Optimality Discussion

8.1 Is This the True Minimum L^2 Error?

The mergeability score ε uses moment-matched $\bar{\boldsymbol{\mu}}, \bar{\boldsymbol{\Sigma}}$ with L^2 -optimised amplitude. The *globally* L^2 -optimal Gaussian would also optimise $\bar{\boldsymbol{\mu}}$ and $\bar{\boldsymbol{\Sigma}}$, satisfying:

$$\sum_i a_i \lambda_i (\boldsymbol{\Sigma}_i + \bar{\boldsymbol{\Sigma}})^{-1} (\bar{\boldsymbol{\mu}} - \boldsymbol{\mu}_i) = 0, \quad (28)$$

where $\lambda_i = \langle G_i, \bar{G} \rangle$ depends on $\bar{\boldsymbol{\mu}}$ and $\bar{\boldsymbol{\Sigma}}$ —a nonlinear system with no general closed-form solution. Thus:

- The score ε is an **upper bound** on the true minimum relative L^2 error.
- In practice, the moment-matched and L^2 -optimal solutions are **very close** when the two covariances are similar (the typical case for merge candidates).

- For exact minimisation when needed, a few iterations of Gauss–Newton starting from the moment-matched solution converge rapidly.
- The error formula (17) is exact for *any* choice of g , so one can always evaluate the true L^2 error once the optimal parameters are found numerically.

8.2 When Does Moment-Matching Equal L^2 -Optimal?

1. **Proportional splats** ($\phi_1 \propto \phi_2$): trivially, $\varepsilon = 0$ for both.
2. **Equal amplitudes and equal covariances** ($a_1 = a_2$, $\Sigma_1 = \Sigma_2$): by symmetry, the optimal mean is $(\mu_1 + \mu_2)/2$, which is also the moment-matched mean. The optimal covariance still differs from the moment-matched one in general, but the difference is typically small.

8.3 Relationship to Existing Merging Criteria

- **Runnalls’ KL bound** [5]: an upper bound on the KL divergence cost of merging. It is $B_{12} = \frac{1}{2}[(w_1 + w_2) \log |\bar{\Sigma}| - w_1 \log |\Sigma_1| - w_2 \log |\Sigma_2|]$, which only measures the covariance mismatch and is blind to amplitude differences. Our L^2 score incorporates amplitudes and the full geometric overlap.
- **ISE-based reduction** [7, 9]: uses $\|f - g\|_{L^2}^2$ directly, as we do. Our contribution is the explicit factorisation into the $\sin^2 \alpha$ form and the amplitude-optimised closed-form expression.
- **Salmond’s method** [6]: merges components with the smallest increase in ISE, equivalent to choosing the pair with the smallest E^* .

8.4 Practical Considerations

Truncation. The Luxar implementation uses a shifted truncated Gaussian $a \cdot s \cdot \max(0, e^{-\mathcal{M}^2/2} - C)$, where $\mathcal{M}$ is the Mahalanobis distance, $s = 1/(1-C)$ is a peak-preserving scale factor, and $C = e^{-T^2/2} \approx 0.011$ for $T = 3$. The tails beyond T standard deviations contribute $< 1.2\%$ of the total energy, so the pure-Gaussian analysis provides an excellent approximation.

Computational cost. For $D = 3$: three Cholesky factorisations of 3×3 matrices, three linear solves, and scalar arithmetic. Total: ~ 200 floating-point operations per pair, negligible compared to the rendering cost.

Batch processing. For culling, precompute $J_i = \pi^{D/2} |\det \mathbf{L}_i|$ and m_i for all splats. The per-pair cost then reduces to one kernel evaluation I_{12} (for the cosine proxy) or four kernel evaluations (for the full score).

9 Summary

1. The L^2 integrated squared error is the unique divergence for which *all* terms in the Gaussian merging error have closed-form expressions.

2. The moment-matched Gaussian (KL-optimal) provides a closed-form approximation whose shape parameters are given by (13)–(14), with L^2 -optimal amplitude from (18).
3. The normalised merging error is $\varepsilon = \sin^2 \alpha$ (Eq. (20)), where α is the angle between the mixture and the template Gaussian in L^2 .
4. For ranking pairs cheaply, the inter-splat cosine (Eq. (22)) requires only one kernel evaluation and is amplitude-independent.
5. For equal isotropic covariances, the cosine reduces to $\exp(-d^2/4\sigma^2)$, depending only on the normalised separation.

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